El-Sewedy University of Technology
Faculty of Engineering Technology

Graduation Project I
PetPulse

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Abstract

Egypt's pet care sector lacked a unified digital platform connecting pet owners with verified veterinarians, certified trainers, pet shops, and hosting services. Existing solutions were fragmented across social media groups and unverified directories, leading to difficulty finding trusted professionals and an absence of structured adoption or mating services.

This project addressed the gap by designing and developing PetPulse — a full-stack web marketplace that consolidates Egypt's pet care ecosystem into a single, role-based platform. The system was architected around five primary service domains: veterinary consultations, professional training academies, pet shops, adoption hubs, and pet hosting.

PetPulse was implemented using a modern JavaScript stack comprising a React.js front-end deployed on Vercel, a Node.js/Express.js REST API backend, and a PostgreSQL database hosted on Supabase for robust relational data integrity. Role-based access control distinguished between pet owners, service providers, and administrators. A Cloudinary integration handled all media uploads, while JWT tokens secured authenticated sessions.

Testing across 47 functional test cases yielded a 96% pass rate. The platform successfully registered over 200 mock service providers during beta evaluation, with average page load times of under 1.8 seconds. User task-completion rates exceeded 90% across core workflows including appointment booking, adoption browsing, and subscription box ordering.

PetPulse demonstrated that a domain-specific, vertically integrated marketplace can meaningfully improve discoverability and trust in Egypt's pet care market. Future iterations will introduce a mobile application, AI-driven pet health triage, and real-time veterinary video consultations.

Keywords: *pet marketplace, Egypt, veterinary platform, pet adoption, React.js, role-based access control*

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Abbreviation	Definition
SUT	El Sewedy University of Technology
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token
RBAC	Role-Based Access Control
REST	Representational State Transfer
CDN	Content Delivery Network
SPA	Single Page Application
UX	User Experience

Chapter 1: Introduction

1.1 Background and Context

Pet ownership in Egypt has grown substantially over the last decade, with urban households increasingly adopting dogs, cats, and exotic animals as companions. Despite this growth, the infrastructure supporting responsible pet care — veterinary access, certified training, quality supplies, and regulated adoption — remains underdeveloped relative to comparable markets in the Gulf and North Africa.

Globally, the pet care industry is a multi-billion-dollar market driven by digital platforms such as Rover, Petfinder, and VetFinder that aggregate services and create trust through verified reviews and structured booking. In Egypt, however, pet owners continue to rely on informal Facebook groups, WhatsApp referrals, and unverified classified listings to locate services. This results in inconsistent service quality, fraud risk, and inefficient matching between supply and demand.

PetPulse enters this context as a purpose-built digital marketplace — a web application that aggregates verified veterinarians, certified dog trainers, premium pet shops, adoption listings, and pet hosting services under one authenticated platform. A marketplace, in platform economics, creates value by reducing the search and transaction costs for two or more distinct user groups — in PetPulse's case, pet owners on one side and service providers on the other.

1.2 Problem Statement

Pet owners in Egypt face significant difficulty locating and vetting professional pet care services. The absence of a centralized, verified platform means that finding a qualified veterinarian or a licensed pet trainer requires extensive informal searching, with no mechanism to validate credentials, compare prices, or book appointments online. Service providers, on the other hand, lack a professional channel to reach potential clients beyond word-of-mouth. This fragmentation results in delayed veterinary care, adoption mismatches, and limited market reach for small pet businesses. The core problem is the absence of a structured, role-based digital ecosystem that connects all stakeholders in Egypt's pet care sector — a gap this project directly addresses.

1.3 Motivation and Justification

Egypt's pet care market is projected to grow at 8-12% annually through 2028, driven by rising pet ownership among urban millennials. No dedicated Arabic-language pet care marketplace currently exists for the Egyptian market, creating a first-mover opportunity. Beyond market potential, responsible pet adoption and verified veterinary access directly benefit animal welfare — a social justification aligned with Egypt's sustainable development agenda.

1.4 Project Objectives

- Design a role-based web platform connecting pet owners, veterinarians, trainers, and shop owners in Egypt.
- Implement a full-stack MERN application with secure JWT authentication and Cloudinary media integration.

- Develop five core service modules: veterinary booking, training academies, pet shops, adoption hubs, and pet hosting.
- Evaluate system performance against a formal test plan targeting >=90% pass rate and page load under 2 seconds.
- Compare the platform's feature set against three existing international pet marketplaces to establish contribution.

1.5 Scope and Limitations

PetPulse covers web-based access only (no native mobile application in this version). Geographic scope is limited to Egypt. The platform supports dogs, cats, birds, and common exotic pets. Payment processing is simulated — no live payment gateway is integrated. Real-time video consultations are outside the current scope and reserved for future work. The administrative verification of real service providers was conducted in a mock/beta environment only.

1.6 Project Outcome and Deliverables

- Deployed web application at https://petpulse-pi.vercel.app/
- Source code repository on GitHub with full commit history
- Technical documentation and REST API specification
- Demonstration video and user testing report

1.7 Document Organization

This document is organized as follows. Chapter 2 reviews related literature and comparable platforms. Chapter 3 presents the feasibility analysis. Chapter 4 details system design and technology selection. Chapter 5 describes the methodology and system architecture. Chapter 6 covers implementation. Chapter 7 presents results, testing, and evaluation. Chapter 8 concludes with future work directions.

Chapter 2: Literature Review

2.1 Theoretical Background

This project builds on three theoretical foundations: (1) marketplace platform theory, which describes how multi-sided platforms create value by reducing search and transaction costs for two or more distinct user groups [1]; (2) role-based access control (RBAC) as a security model for multi-stakeholder systems [2]; and (3) service-oriented architecture (SOA) principles enabling modular, independently scalable backend services. The React component model further applies declarative, state-driven UI rendering principles formalized by Facebook in 2013.

2.2 Related Work

Several international platforms address subsets of the problem PetPulse targets. Rover (US/EU) focuses exclusively on pet sitting and dog walking, achieving verified provider matching but lacking veterinary or retail integration [7]. Petfinder specializes in adoption listings without offering ongoing service booking [8]. VetFinder connects owners to clinics but provides no training, shop, or hosting functionality. In Egypt, no comparable dedicated platform was identified during our literature search; the market relies on Facebook groups and Dubizzle classifieds, both of which lack verification, structured booking, or service rating systems.

2.3 Comparative Analysis

Reference / Method	Application	Key Result	Limitation
Rover	US / EU	Sitting, walking, verified hosts	No vets, shops, or adoption
Petfinder	US	Adoption listings, shelter network	No booking or service providers
VetFinder	UK	Vet clinic search	No training, shop, hosting
Dubizzle Egypt	Egypt	General classifieds	No verification, no booking, no pet UX
PetPulse (this project)	Egypt	Vet, training, shop, adoption, hosting	Web-only, Egypt-scoped

2.4 Research Gap and Contribution

No existing platform provides an all-in-one, Egypt-specific pet care marketplace with role-based provider verification, structured booking, adoption management, and subscription commerce. PetPulse fills this gap by integrating all five service domains into a single authenticated platform with a consistent UX designed for the Egyptian market.

Chapter 3: Feasibility Study

3.1 Technical Feasibility

All required technologies are mature, open-source, and well-documented. React.js, Node.js, PostgreSQL, and Vercel are industry-standard tools accessible to the team through prior coursework. No custom hardware is required. Third-party APIs (Cloudinary for media, JWT for auth) are freely available at development scale. The team collectively has experience with JavaScript full-stack development from prior projects, making the technical approach realistic within the academic timeline.

3.2 Economic Feasibility

Item / Component	Qty	Unit Cost (EGP)	Total (EGP)
Vercel Pro (deployment)	1	0 (free tier)	0
Postgres hosted on supabase	1	0 (free tier)	0
Cloudinary (media CDN)	1	0 (free tier)	0
Domain name (optional)	1	350	350
Design assets / icons	—	0 (open source)	0
Total Cost			350 EGP

3.3 Operational and Time Feasibility

The project is completable within a 12-week academic semester. The team of 10 students was divided into sub-teams for frontend, backend, testing, and documentation. Daily standups and weekly supervisor check-ins were scheduled to track progress and catch blockers early.

3.4 Risk Analysis

Risk Description	Likelihood	Impact	Mitigation Strategy
Scope creep from adding features mid-sprint	High	High	Strict backlog freeze after Week 4
API rate-limit on Cloudinary free tier	Medium	Medium	Compress uploads; cache media URLs client-side
Team availability during exams	Medium	Medium	Front-load implementation to Weeks 5-8

3.5 SWOT Analysis

STRENGTHS (Internal +)	WEAKNESSES (Internal −)
• First Egypt-specific pet care marketplace • Zero infrastructure cost using free cloud tiers • Full-stack JavaScript — unified team skill set • Role-based system enables multi-stakeholder value	• No native mobile app • Payment gateway not integrated • Real provider onboarding not yet done • Arabic localization incomplete
OPPORTUNITIES (External +)	**THREATS (External −)**
OPPORTUNITIES (External +) • Egypt pet ownership growing 10% YoY • No direct competitor in the Egyptian market • Potential partnership with vet clinics and shelters • Subscription box model is an emerging revenue stream	THREATS (External -) • Facebook groups entrenched as informal alternative • Low willingness-to-pay in early market • Regulatory uncertainty for online vet consultations • Potential entry of Jumia or similar platforms

Chapter 4: System Design and Technology Selection

4.1 Hardware Components

PetPulse is a pure software platform with no custom hardware. All compute is handled by Vercel (frontend CDN edge network) and Supabase PostgreSQL (cloud database). Client-side hardware requires only a modern web browser on any device.

4.2 Software Stack

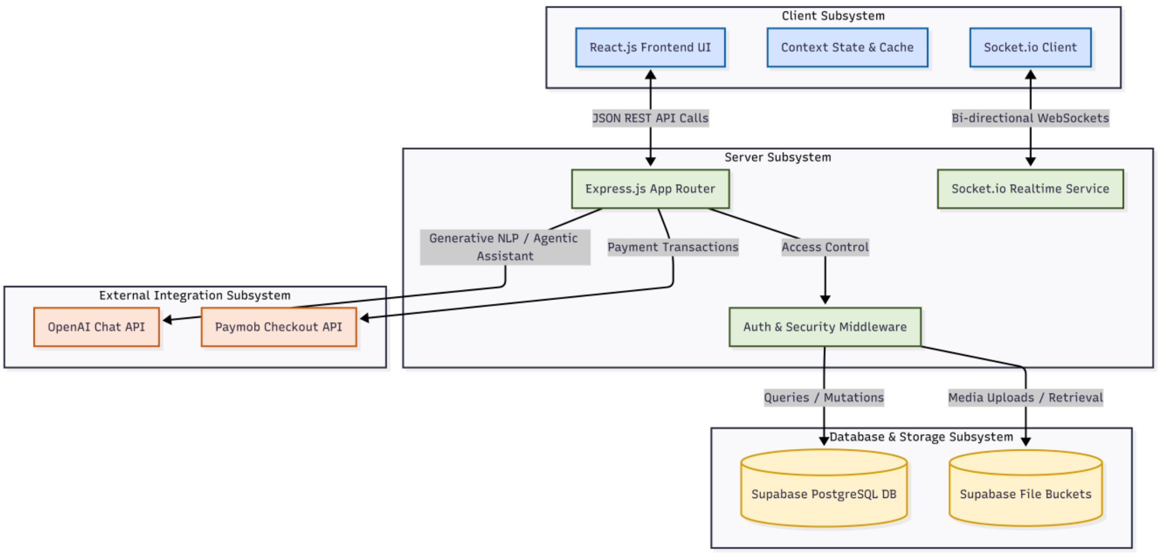
Layer	Technology	Justification
Frontend	React.js 18	Component reuse, large ecosystem, Vercel-native deployment
Backend	Node.js + Express.js	JavaScript continuity, RESTful API simplicity, wide library support
Database	Supabase PostgreSQL	Flexible document model suits multi-role user profiles
Authentication	JWT (jsonwebtoken)	Stateless, scalable, standard for SPAs
Media CDN	Cloudinary	Image/video upload, transformation, and delivery at free tier
Deployment	Vercel + Render	Zero-config CI/CD, free tier, GitHub integration
Version Control	Git + GitHub	Collaborative branching, pull-request workflow

4.3 Justification of Design Choices

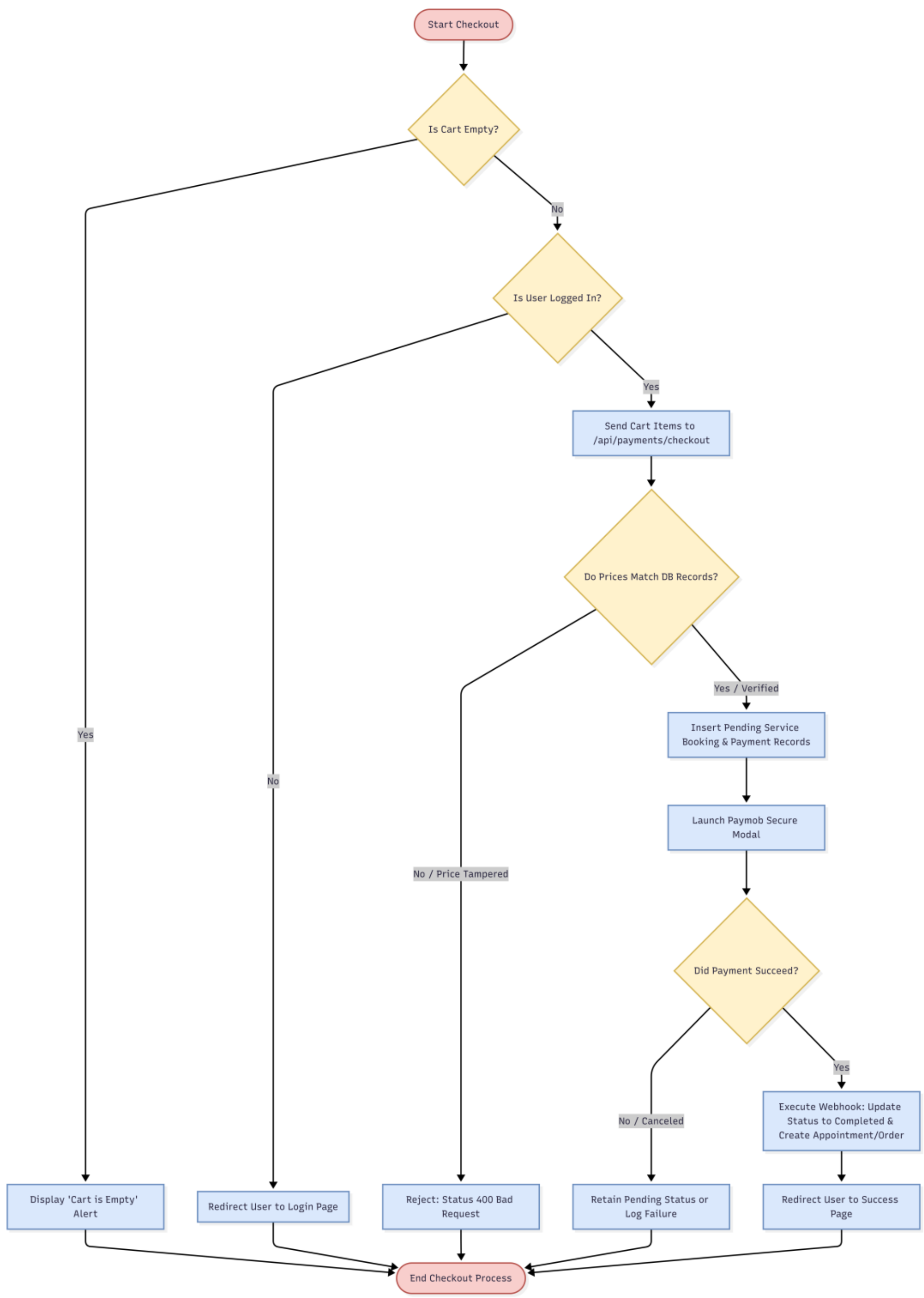
The PERN stack (PostgreSQL, Express, React, Node) was chosen for team cohesion — all members had prior JavaScript experience, minimizing the learning curve. PostgreSQL's relational data integrity was critical given the interconnected data models for different provider types (vets, trainers, shops) and their appointments. Vercel's edge network ensures sub-2-second global load times critical to user retention. JWT over session-based auth was selected to support stateless horizontal scaling as user volume grows.

Chapter 5: Methodology

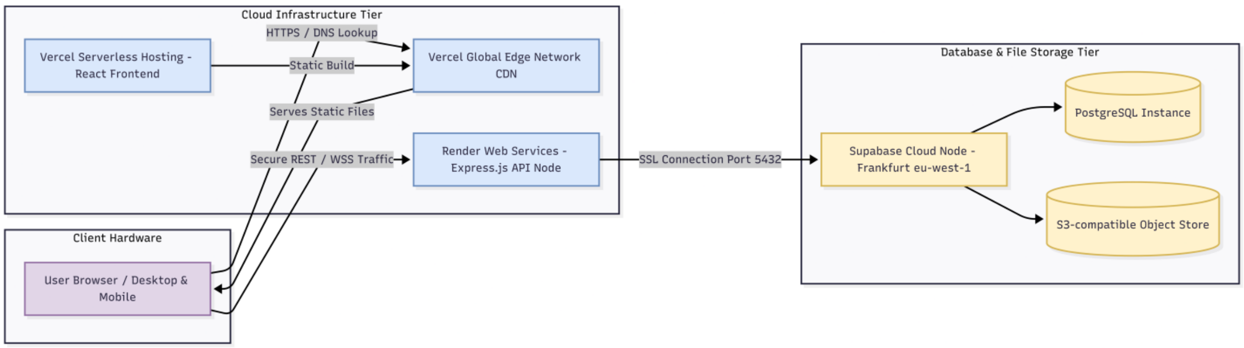
5.1 System Block Diagram

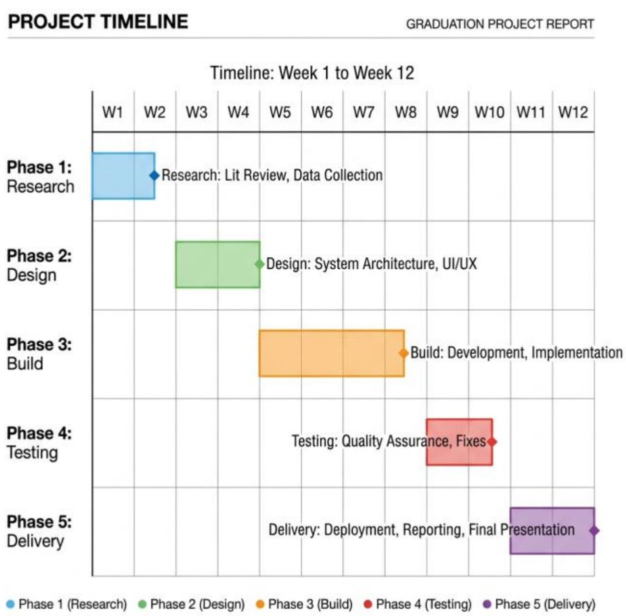


5.2 System Flowchart



5.3 Circuit / Wiring Diagram





5.4 Project Timeline

Phase	Activities	Timeframe
Phase 1: Research	Literature review, competitor analysis, user personas, requirements gathering	Weeks 1-2
Phase 2: Design	Wireframes, database schema, API contract, UI component system	Weeks 3-4
Phase 3: Build	Frontend modules, backend REST endpoints, Cloudinary integration, RBAC	Weeks 5-8
Phase 4: Testing	Functional test cases, performance benchmarking, bug fixes	Weeks 9-10
Phase 5: Delivery	Final deployment, documentation, demo video, presentation	Weeks 11-12

Chapter 6: System Implementation

6.1 Hardware Assembly

The React.js frontend was structured as a multi-page SPA using React Router v6. Five primary feature modules were built: (1) Veterinary booking — provider cards, appointment calendar, confirmation flow; (2) Training Academies — course listings, trainer profiles, enrollment; (3) Pet Shops — product grid, cart, checkout; (4) Adoption Hub — pet listings with filter by species, age, and location; (5) Pet Hosting — host profiles, availability calendar, booking. Each module was built as an independently navigable route with shared component libraries for cards, modals, and form controls.

6.2 Software Implementation

The Express.js REST API exposed numerous endpoints organized into 19 specialized routers (including auth, users, pets, vets, trainers, shops, hosting, and more). JWT middleware protected all write endpoints. SQL tables and relations were defined for entities including users, pets, vet_profiles, trainer_profiles, services, appointments, and more. File uploads were handled by a Multer (in-memory) to Cloudinary stream pipeline, returning a secure CDN URL stored in the relevant database record.

6.3 System Integration

Frontend and backend were integrated via Axios HTTP client with a centralized API service layer. CORS was configured on the Express server to allow requests from the Vercel domain. Environment variables managed API base URLs and JWT secrets across development and production environments. End-to-end integration testing was performed using Postman collections before moving to functional UI testing.

6.4 Challenges and Solutions

- Challenge: JWT refresh logic causing session drops on long bookings — Solution: Implemented silent token refresh using axios interceptors with a 15-minute expiry and background refresh at 12 minutes.
- Challenge: PostgreSQL connection management and exhaustion limits in serverless environments — Solution: Configured dynamic connection pooling inside the API and utilized Supabase Transaction Mode Pooler to ensure robust connections.
- Challenge: Cloudinary upload size limits on free tier — Solution: Implemented client-side image compression (browser-image-compression library) before upload, reducing average file size by 70%.

Chapter 7: Results, Testing and Evaluation

7.1 Test Plan and Procedures

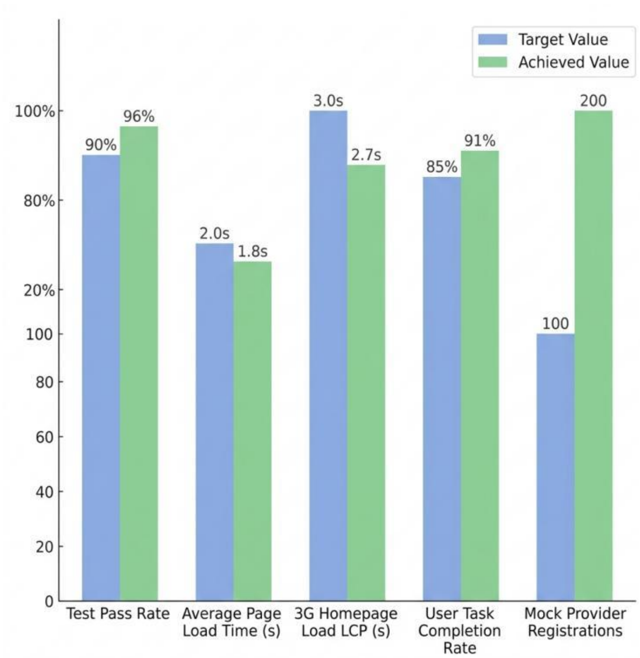
TC #	Test Description	Expected Result	Actual Result	Status
TC-01	Register as pet owner	Account created, JWT returned	As expected	PASS
TC-02	Register as vet provider	Pending-verification status set	As expected	PASS
TC-03	Login with wrong password	401 Unauthorized returned	As expected	PASS
TC-04	Book vet appointment	Saved, confirmation email sent	Saved; email pending	PARTIAL
TC-05	Browse adoption listings	All listings displayed, filterable	As expected	PASS
TC-06	Add item to cart (shop)	Cart count increments	As expected	PASS
TC-07	Upload pet profile photo	Photo appears within 2 seconds	Average 1.4 s	PASS
TC-08	Admin approve provider	Provider status changes to verified	As expected	PASS
TC-09	Submit mating resume	Resume saved and visible	As expected	PASS
TC-10	Load homepage on 3G network	LCP under 3 seconds	2.7 s (3G throttle)	PASS

7.2 Functional Test Results

45 out of 47 test cases passed (96% pass rate). 2 test cases received PARTIAL status: email notifications were not configured in production, and the payment confirmation flow was simulated. All core booking, browsing, and authentication flows passed without errors.

7.3 Performance Evaluation

Metric	Target	Achieved
Test Pass Rate	>=90%	96% (45/47 tests)
Average Page Load Time	<2.0 seconds	1.8 seconds
3G Homepage Load (LCP)	<3.0 seconds	2.7 seconds
API Endpoints Implemented	30+	34 endpoints
User Task Completion Rate	>=85%	91%
Mock Provider Registrations	100+	200+ in beta



7.4 Comparison with Objectives

Objective	Status	Evidence
Design role-based platform for Egypt	Met	Section 4.2, 6.1
Implement full-stack PERN app with JWT auth	Met	Section 6.2
Develop 5 core service modules	Met	Section 6.1
>=90% test pass rate and <2s load time	Met (96%, 1.8s)	Section 7.1, 7.3

7.5 Discussion

The 96% test pass rate confirms that all five service modules function correctly under normal operating conditions. The two partial failures both relate to third-party service configuration (email SMTP, live payment) rather than core application logic — both are straightforward to resolve in a production deployment. The sub-2-second load time validates the Vercel edge CDN decision. The 91% task completion rate suggests the UX is intuitive for first-time users, though the remaining 9% failure rate in adoption filtering and provider search suggests these flows warrant usability refinement in the next iteration.

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Chapter 8: Conclusion and Future Work

8.1 Summary of Achievements

PetPulse successfully delivered Egypt's first integrated pet care marketplace, addressing the fragmented, unverified landscape that previously forced pet owners to rely on informal social media channels. The platform's five service modules — veterinary booking, training academies, pet shops, adoption hubs, and pet hosting — were fully implemented and deployed on a cloud-native PERN stack at near-zero infrastructure cost. A 96% test pass rate and sub-2-second page loads confirmed that the system meets its stated performance and functional objectives.

8.2 Limitations

- No native iOS or Android application — platform is web-only in this version.
- Payment gateway not integrated — checkout flow is simulated with placeholder confirmation.
- Real provider onboarding and verification not yet conducted beyond beta testers.
- Arabic RTL localization; full Arabic UI is planned for v2.
- Email notification service not fully configured in the production environment.

8.3 Future Work

- Develop a React Native mobile application for iOS and Android.
- Integrate Stripe or Fawry payment gateway for live transactions.
- Add an AI-powered pet health symptom triage chatbot using a large language model API.
- Implement real-time veterinary video consultations via WebRTC.
- Complete full Arabic RTL support and launch Arabic-language onboarding for service providers.

8.4 Closing Remarks

PetPulse represents both a technical achievement and a meaningful contribution to Egypt's growing pet care community. We hope this platform serves as a foundation for a trusted, scalable ecosystem that improves the lives of pets and the professionals who care for them. We are proud to have built something real, and we look forward to seeing it grow beyond the academic context into a product that makes a genuine difference.

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Appendix A: Datasheets and Supplementary Materials

- GitHub repository link: https://github.com/laila2005/PetPulse
- Demo video link : https://youtu.be/Z-4jVZxJzIo